

Supplementary Material

RareAgentBench: An Agent-Native Benchmark for Rare-Disease Diagnosis

Anonymous submission

This document is the supplementary material for *RareAgentBench*. It is self-contained: each appendix restates the setup it needs, so it can be read without the main paper. It documents (A) the data layers and their construction, (B) the backbone configurations, (C) the agent fairness matrix and adapter patches, (D) per-baseline reproduction against published claims, (E) the temporal-holdout protocol, (F) the pre-registered hypotheses and their family-wise test results, (G) the full cost accounting, (H) a MIMIC-IV note probe with a 24-cell agent matrix, and (I) bootstrap and reproducibility notes. Unless stated otherwise, the primary metric is cross-ontology Recall@1 (R@1) with an *attempted* denominator: every empty output, parser error, and timeout is counted as a miss, never dropped. Disease predictions are matched by exact identifier, then accepted OMIM-ORPHA mappings, then Orphanet name/synonym match with a RapidFuzz threshold of 90.

A Data Layers and Construction

RareAgentBench draws on a source pool of 84,450 candidate instances from five sources, organized into four layers that vary evidence structure, clinical realism, scale, and contamination risk. Table 1 summarizes the layers as evaluated in this release; the per-layer counts are the released sizes and do not sum to the candidate pool.

Table 1: Data layers. “Input” is the evidence an agent receives; “Gold” is the reference label source. Counts are released-pool sizes; per-experiment evaluated N is reported per cell in the main paper.

Layer	Source (count)	Input	Gold
L1	Phenopacket Store (10,051)	curated HPO + variants	OMIM/ORPHA
L1	RareBench-HF (1,122)	sparse HPO list	ORPHA
L2	MIMIC-IV notes (416)	masked discharge summary	ICD→ORPHA (exact)
L3	RareArena RDS (8,562)	free-text vignette	ORPHA
L4	PMC-OA holdout (198)	post-cutoff case text	ORPHA (adjudicated)

L1 Phenotype backbone. Phenopacket Store contributes standardized cases with curated HPO terms and, where available, structured variants; RareBench-HF contributes 1,122 cases for comparability with established rare-disease LLM

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evaluation. These are the HPO-input layers on which classical/offline baselines (LIRICAL, VC-RDAGent) can run, since those systems require a structured phenotype list.

L2 Real-EHR noise. A discharge-note probe over MIMIC-IV v3.1, frozen at 416 cases / 68 diseases (sha256 67d156aa...). Candidate admissions are identified through ICD-10-ORPHA mappings, excluding Orphadata categories marked non-rare in Europe. Because MIMIC-IV is credentialed PhysioNet data under a Data Use Agreement, no row-level text or identifiers are redistributed; only aggregate metrics and hashes leave the authorized machine. The note-based probe built from this source is documented in Appendix H.

L3 Free-text scale. RareArena RDS contributes 8,562 free-text records in the current release, derived from PubMed Central case reports; the RDC confirmation view is not included in this release. We count these as task instances rather than unique patients.

L4 Temporal holdout. See Appendix E for the full post-cutoff extraction and adjudication protocol.

B Backbone Configurations

Every LLM-based system is evaluated, where technically applicable, with four backbones spanning the observed cost-capability range. Requests are routed through a common OpenAI-compatible gateway to give a single API surface, billing record, and generation log. We use dated model identifiers, not moving “latest” aliases. Table 2 lists the configurations.

Table 2: Backbones evaluated, with pricing and reasoning configuration. Primary comparisons use the lowest reasoning setting each endpoint reliably supports, to reduce backbone-side test-time compute as a confound when measuring scaffold effects and to keep multi-call evaluation tractable.

Role	Model	Price (\$/M in / out)	Reasoning
Open-cheap	DeepSeek V4-Flash	0.28 / 0.42	default light
Open-frontier	DeepSeek V4-Pro	0.55 / 2.19	disabled
Mid (primary)	Gemini 3 Flash	0.50 / 3.00	thinking off
Frontier	GPT-5	1.25 / 10.00	effort=minimal

Settings held constant across all (agent, backbone) cells: temperature 0.0; seed 42 where the SDK exposes one; per-call timeout 600–1200 s; a retry policy with exponential backoff capped at three attempts; and `max_tokens` at each adapter’s published default (2K–6K).

Constrained-reasoning regime. For the frozen primary runs, GPT-5 uses `reasoning_effort=minimal` and V4-Pro uses reasoning disabled. This lowest-reasoning setting reduces backbone-side test-time compute as a confound when measuring scaffold effects and keeps multi-call evaluation tractable. All conclusions about GPT-5 and V4-Pro apply specifically to this constrained-reasoning regime; a reasoning-enabled single-LLM control is examined in Appendix F (H6). Failure handling is fixed before aggregation: empty outputs, parser errors, and timeouts count as unsuccessful attempts and remain in the denominator; a cell is marked unavailable only when the adapter–backbone pair cannot execute the required interaction protocol.

C Agent Fairness Matrix and Adapters

The diagnostic systems in Table 3 have mutually incompatible dependencies. Each runs in an isolated runtime behind a common adapter interface that receives a canonical case, exposes only the evidence permitted by the current capability and pass, converts it to the system’s native input, invokes the isolated process, and returns a prediction log with ranked diagnoses, extracted HPO terms, traces, token usage, cost, latency, and a status code. Adapter modifications are restricted to transport, input projection, output parsing, and instrumentation—never to prompts, tool-selection policies, stopping rules, retrieval corpora, or orchestration logic. Table 3 records each system’s native input, adapter strategy, per-case LLM calls, and license.

Audit properties and state isolation. Every run receipt records the canonical case identifier, adapter version, model identifier, configuration hash, subprocess command, random seed, and termination status. A common logging layer aggregates per-call token usage, latency, and billed cost into a case-level receipt; API costs are exact when returned by the gateway and are otherwise estimated from token counts and versioned prices (labeled accordingly, $\leq 5\%$ band). Process isolation contains dependency, parser, retrieval, and state failures within the affected system. Interpreter startup adds roughly 0.5–2 s per case, measured separately from end-to-end latency; system-specific output directories are cleared between cases to prevent cross-case state leakage.

D Per-Baseline Reproduction

Before integrating each system we reproduced its published headline against a small pilot, to confirm the adapter preserves native behavior. Table 4 reports the outcome. Six of nine systems reproduce within ± 5 points; the two that underperform (MAI-DxO, DeepRare-P2) do so for a documented input-distribution reason—both were built for narrative-rich or VCF-bearing inputs and are here given HPO lists or brief vignettes—rather than an implementation error.

The DeepRare P2→P3 lift (+20 points from adding a structured variant block) matches the same lift obtained by a plain single-LLM control given the same variants, which is why we report the genotype-channel benefit as general rather than DeepRare-specific.

E Temporal-Holdout Protocol

The L4 holdout targets contamination by drawing cases published after the backbones’ training cutoff. From 2,401 PubMed Central candidates we obtain 1,433 mapped records with a fixed extraction-and-mapping pipeline, screen the top-ranked 250, remove 52 non-post-cutoff reports, and retain 198 cases satisfying four criteria: a definitive diagnosis; faithful phenotype extraction; verified publication after the pre-specified cutoff; and confirmation that the target condition is rare. The holdout is evaluated once, under the frozen agent set, matching rules, and analysis plan; no holdout result is used to revise prompts, adapters, or thresholds.

Difficulty-matched pre/post comparison. A naive post-cutoff score is confounded by difficulty, so we build a pre-cutoff set with the identical pipeline (same source, query, extraction, mapping, and gold-verification), changing only the publication window. On the clean-gold subset, pooled single-pass R@1 does not drop across the cutoff; if anything it is marginally higher on post-cutoff cases, consistent with newer reports being slightly clearer. We therefore report “no detectable post-cutoff degradation” after difficulty matching, rather than a claim of contamination-freeness: the holdout is not guaranteed disjoint from every development source, so it bounds contamination from a second angle rather than eliminating it.

F Pre-Registered Hypotheses and Family-Wise Tests

Before temporal-holdout unblinding we froze hypotheses H1–H11, the ablation set, the staged compute budget, cell-level stopping rules, primary metrics, the Holm–Bonferroni correction, and the LLM-judge agreement protocol. The frozen protocol, a timestamped repository commit, and an anonymized read-only registration accompany the release. Table 5 lists the hypotheses; Table 6 reports the family-wise test over the testable subset.

Five of the six testable hypotheses survive the correction. H10 (faithfulness–accuracy decoupling) is reported as exploratory and judge-dependent: on identical repaired traces the faithfulness↔ accuracy Spearman correlation is $\rho = 0.457$ under a same-family (Gemini) judge and $\rho = 0.640$ under a cross-family (Claude) judge, straddling the pre-registered $\rho < 0.5$ boundary. Because the two judges place the correlation on opposite sides of the threshold, we report H10 as exploratory and judge-dependent rather than as a confirmed effect.

Three of the surviving hypotheses are shown directly. Figure 1 visualizes H2: adding a structured variant block to the phenotype-only input lifts R@1 for any system that ingests it. Figure 2 visualizes H8: R@1 peaks at 16–30 HPO terms per case and falls off on either side. Figure 3 visualizes H7:

Table 3: Agent fairness matrix. “Calls/case” is the number of backbone invocations per case. Any deviation forced by an unavailable endpoint is declared here; no prompts or orchestration logic were altered.

System	Native input	Adapter strategy	Calls/case	License
MDAgents	MCQA prompt	rank-top-5 reformulation; regex-parse moderator; remove hard-coded model	7–47	None
MedAgents	MCQA prompt	bypass MCQA runner; 3 experts + Chief synthesis	~10	None
AgentClinic	OSCE dialogue	synthetic OSCE loop; 2nd call for ranks 2–5	~45	MIT
MAI-DxO	Panel orchestration	router + variant factory; ≥ 2 iterations	10–50	MIT port
DeepRare	Phenopacket + text	local-embedding, no-web environment	20–40	CC BY-NC 4.0
RDMA	EHR free-text	entity extraction; Pillar 1 only	1–3	None
VC-RDAGENT	HPO list	offline Stage 1 (IC + Poincaré + freq-LR); Stage 2 opt-in	0 (Stage 1)	None
LIRICAL	GA4GH phenopacket	subprocess <code>lirical phenopacket</code> ; parse TSV	0	Apache 2.0

Table 4: Per-baseline reproduction on an $n=50$ pilot. “Ours” is our pilot point estimate; “Paper” is the originally reported figure on that system’s own evaluation set. Discrepancies are attributed to input-distribution mismatch, documented per row.

System	Ours (pilot)	Paper	Note
MDAgents	R@1 0.34	0.31–0.39	rank-top-5 reformulation; ± 5 pt
MedAgents	R@1 0.36	0.32	3 experts + Chief; ± 5 pt
AgentClinic	R@1 0.30	0.28	synthetic OSCE; ± 5 pt
MAI-DxO	R@1 0.22	0.45	paper input = NEJM narratives; ours = HPO list
DeepRare (P2)	R@1 0.22	0.71	paper input includes VCF; P2 excludes variants
DeepRare (P3)	R@1 0.42	0.71	structured-variant block, web disabled
RDMA	F1 0.39	0.42	Pillar-1 extraction only
VC-RDAGENT	R@1 0.28	0.27	Stage-1 offline, 0 LLM calls
LIRICAL	R@1 0.40	~0.42	pilot; 0.47 on the full frozen matrix

the per-organ R@1 ordering is shared across LLM systems, and the classical baselines invert it on the nervous-system and head/neck strata where LLMs are weakest.

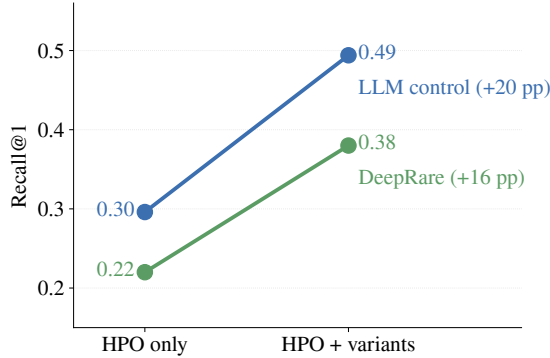


Figure 1: H2 genotype channel. R@1 when the same cases are scored on phenotype terms alone versus phenotype terms plus a structured variant block, for the single-LLM control and for DeepRare; both gain about 16–20 points.

G Cost Accounting

The evaluation comprises 97,985 predictions at a total API cost of \$297.89, or \$0.0030 per prediction including offline runs. Per-prediction cost spans more than an order of magnitude across backbones (Table 7, Figure 4): GPT-5 minimal is the most expensive without a consistent accuracy advantage, while V4-Flash is more than an order of magnitude cheaper.

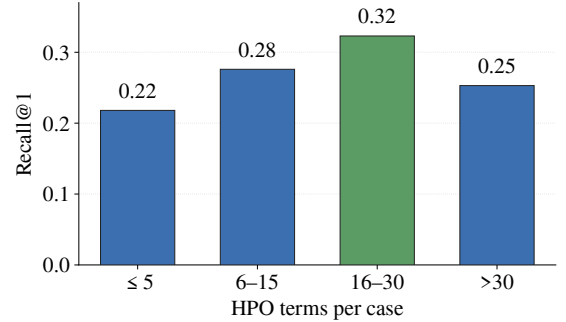


Figure 2: H8 phenotype density. Pooled R@1 by the number of HPO terms per case; the 16–30 band (highlighted) is the peak, an inverted-U consistent with too few terms underspecifying and too many diluting the query.

Offline/classical baselines incur no API cost.

H MIMIC-IV Note Probe

The MIMIC-IV-Note probe uses the *presentation* portion of a discharge summary (chief complaint, history, exam, labs, imaging) as input, truncated before any diagnosis-revealing section, with the gold disease name masked. This avoids the answer-naming shortcut present when an ICD long-title is exposed. The gold label remains code-derived (ICD→Orphanet exact mapping), which we flag as a code-supervision caveat; this probe is the MIMIC-IV-Note column of Table 1 in the main paper.

Table 5: Pre-registered hypotheses. H3/H5/H6/H9/H11 are not part of the confirmatory family-wise test reported in Table 6: H3 is a bounded-difference (equivalence-style) check, H5/H9 require data layers not included in this release (Chinese PUMCH corpus; pedigree-bearing family cases), H6 is a descriptive reasoning-mode ablation, and H11 was retired.

#	Statement	Test
H1	classical > scaffolded LLM on super-rare ($<1/10^6$)	2-prop z
H2	genotype channel (P3) $\geq +10$ pp over P2	paired z
H3	post- vs pre-cutoff R@1 differ by ≤ 5 pp	diff of props
H4	scaffolding helps more on ≥ 4 -organ vs ≤ 1 -organ cases	diff-in-diff
H5	Chinese (PUMCH) R@1 ≥ 5 pp below English (deferred)	paired z
H6	GPT-5 medium better calibrated than minimal	paired ECE
H7	cross-agent specialty rank $\rho \geq 0.6$	Spearman ρ
H8	inverted-U: R@1 at 16–30 HPO terms $>$ at ≤ 5	2-prop z
H9	family-aware $\geq +10$ pp on AR/XL cases (deferred)	paired z
H10	faithfulness– and accuracy-rank decouple ($\rho < 0.5$)	Spearman ρ

Table 6: Holm–Bonferroni family-wise correction ($\alpha=.05$, one-sided in the predicted direction) over the six testable hypotheses. Statistic is a z -value for H1/H8/H2/H4 and a Spearman ρ for H7/H10.

#	Statistic	Holm-adj. p	Survives
H1	$z = 17.5$	$2.2e-68$	yes
H8	$z = 12.6$	$7.8e-36$	yes
H2	$z = 6.4$	$3.0e-10$	yes
H7	$\rho = 0.92$	$1.6e-03$	yes
H4	$z = 2.6$	$9.0e-03$	yes
H10	$\rho = 0.35$	$3.7e-02$	no (exploratory)

Table 7: Per-prediction cost by backbone, and the multiple relative to the cheapest backbone (V4-Flash). Offline/classical baselines incur no API cost.

Backbone	\$/prediction	$\times$ V4-Flash
DS V4-Flash	0.00032	1.0
DS V4-Pro (reasoning-off)	0.00098	3.1
Gemini 3 Flash	0.00341	10.7
GPT-5 minimal	0.00825	25.8
LIRICAL / VC-RDAgent	0.00000	—

Subset construction. The deterministic funnel is: 150,033 admissions with any ICD–ORPHA mapping $\rightarrow$ 107,144 exact-mapped only (dropping 37,872 narrower-to-broader and 5,017 broader-to-narrower approximate maps) $\rightarrow$ 18,392 genuinely rare gold $\rightarrow$ 2,325 where the rare disease is the admission’s principal diagnosis $\rightarrow$ 1,255 with a usable discharge summary (117 diseases, sha256 f5d9b437...) $\rightarrow$ 1,083 after dropping 172 prior-known mentions (112 diseases) $\rightarrow$ 855 in which the gold name never appears verbatim (class A) $\rightarrow$ 687 after the strict Orphanet synonym/abbreviation screen (105 diseases) $\rightarrow$ 416 restricted to diseases carrying Orphanet disease-level HPO annotation (68 diseases, sha256 67d156aa...), the frozen scored set. Prior-known detection uses a ± 45 -character window around each [MASKED_DIAGNOSIS] token for history cues (“history of”, “hx”, “known”, “s/p”, “diagnosed with”). The syn-

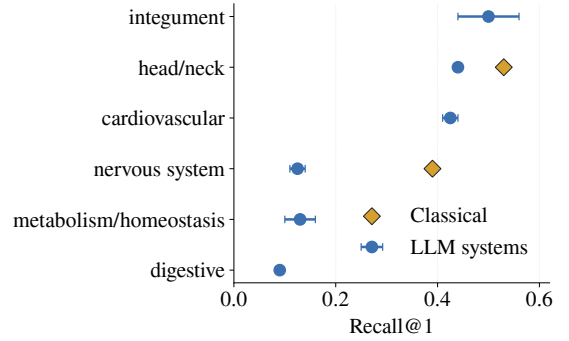


Figure 3: H7 specialty blind spots. R@1 by the case’s modal HPO organ system. Markers are the LLM systems (range shown as a whisker); diamonds mark the classical baseline where it inverts the LLM ordering (nervous system, head/neck).

onym screen matters: 19.6% of the nominally clean class-A notes still contained a gold synonym or abbreviation that the crossmap-aware scorer credits as a hit. Truncation alone left the gold name verbatim in only 2,776/78,166 (3.55%) of notes, all subsequently masked; residual gold in model_input is 0 by programmatic assertion over the full set. The funnel is itself a finding: only $\sim 1,255$ admissions have “a true rare disease as the principal diagnosis with a usable note,” quantifying the validity limit of ICD-derived gold. A parallel capped line (≤ 10 cases per disease, 359 cases / 105 diseases) is used where distribution balance matters more than case count; 416 and 359 are both subsets of the 687 uncapped strict-A set.

Contamination control (presentation vs. full note). Running an unscaffolded LLM on the same 416 cases twice—once on the full, un-truncated, un-masked note and once on the masked presentation—isolates the answer-naming shortcut. R@1 drops on all four backbones: V4-Pro 0.500 $\rightarrow$ 0.365 (−27%), V4-Flash 0.553 $\rightarrow$ 0.353 (−36%), GPT-5 0.425 $\rightarrow$ 0.363 (−15%), Gemini 0.452 $\rightarrow$ 0.373 (−18%); confidence intervals do not overlap. Weaker models lose more, so the shortcut not only inflates absolute scores but compresses and

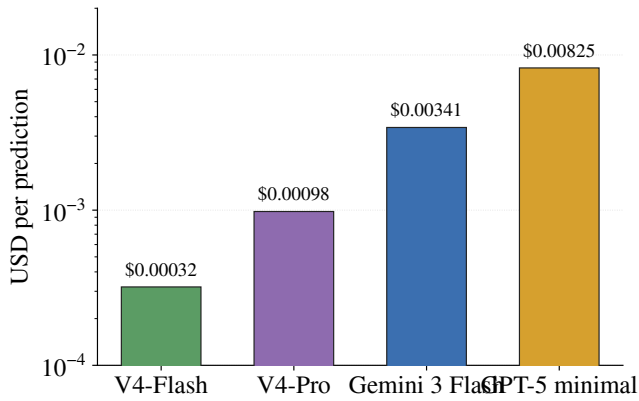


Figure 4: Per-prediction API cost by backbone on a log scale. The cost ordering does not induce a matching accuracy ordering (main paper, Figure 2b); classical/offline baselines incur no API cost and are omitted from the log axis.

reorders the model ranking.

24-cell agent matrix. On the 416-case probe we ran six agent families across four backbones (Table 8, mean micro-R@1 over backbones). The scaffolding picture from the main HPO datasets replicates: the unscaffolded control ties the best light-coordination scaffolds (MedAgents, MDAgents) within overlapping intervals and far exceeds the heavier-orchestration systems, which collapse. This is the second dataset behind the main-paper scaffolding figure. Every one of the 24 cells is complete at 416/416; with the denominator fixed at 416, all failures, timeouts, parser errors and empty-but-successful responses count as misses (1,118 of 9,984 predictions, 11.2%). Per-backbone cells are in Table 1 of the main paper.

Table 8: MIMIC-IV note probe: mean micro-R@1 over four backbones (denominator 416; all failures counted as misses). Scoring is unified across all cells.

Agent	Mean R@1
LLM control (no scaffold)	0.364
MDAgents	0.367
MedAgents	0.350
AgentClinic	0.140
DeepRare	0.011
MAI-DxO	0.003

Honest boundaries. Gold is code-derived, not independent clinical adjudication; note coverage is partial, so a selection bias between admissions with and without a discharge summary cannot be excluded; and an organ-system stratification here is disease-level (from the gold disease’s Orphanet phenotype list), not case-level, so it aligns structurally but not identically with the other datasets; the 271 cases / 37 diseases dropped at the HPO step are an annotation-coverage artifact of Orphanet `en_product4` rather than a quality filter, but they do narrow disease diversity. Finally, the screen is name-

based and case-level: a disease may remain inferable from a characteristic treatment or laboratory pattern. These caveats bound what the MIMIC-IV-Note column supports.

I Bootstrap and Reproducibility Notes

All reported point estimates ship with bootstrap 95% confidence intervals (case-level bootstrap; percentile method). Several small per-agent and per-backbone gaps fall within overlapping intervals and are not treated as definitive rankings. Every confirmatory cell ships a reproducibility receipt containing the run identifier, request identifiers when available, configuration hash, denominator, failure breakdown, and cost. For LLM-judge evaluation of free-text traces, we replay identical frozen traces across judges from different model families, report judge-conditioned results rather than merged scores, and chunk traces longer than 5,000 characters into overlapping 3,000-character windows with 500-character overlap to prevent silent truncation. Deterministic scoring is primary wherever a verifiable target exists; subjective trace scores are reported as exploratory and, for consequential claims, audited against a human-rated subset.